

MethylPhys CPG - Operations Manual, chain v3

Development build, not commissioned; not a diagnostic test. Generated 2026-10-03 by manual/build_manual_v3.py from the files named at the start of each part. The procedure is sop/MethylPhys_CPG_SOP_v3.md.

Part 1. Operations - chain v3, neutrophils

Source: manual/OM_v3_neutrophil_chain.md

Operations Manual — chain v3, neutrophils (operator chapter)

Build: development v3 (2026-10-01), not commissioned. The procedure is in sop/MethylPhys_CPG_SOP_v3.md; this chapter covers running it and reading the output. The PDF manual in this folder (MethylPhys_CPG_Operations_Manual.pdf) is built from this chapter, the generated chain sequence and the toolkit list by build_manual_v3.py. The class-floor engine (v2) and its manual were retired on 2026-10-03 and are archived privately.

Before a run

1. Use EPIC v1 IDAT pairs (Grn and Red), with declared sex and age.
2. Put at least 3 healthy reference specimens on the same slide or batch, run the same way, so whole-blood readings can be tared.
3. Use Python 3.11 with methylprep 1.7.1. The repository includes the frozen files under chain/Runtime Matrices/Met_A_Floors/.

Run

python run_sample.py --grn <Grn> --red <Red> --engine v3 --specimen "whole blood" --array-type EPIC_v1 --sex <F|M> --age <years> --id <id> --out <id>.html Then run each whole-blood specimen again with --slide-ref-A a1,a2,a3, where those are the references' untared A values. The batch runner chain_tests/run_chain_acceptance.py does both passes.

Reading the report

field	meaning
Stage 0 verdict	PROCEED / PROCEED_WITH_PENALTY / QUARANTINE (QUARANTINE produces no reading)
Composition	the 8 blood groups. Neutrophils must be ≥ 50 % for a reading
Met-A	isolated cells: the reading. Whole blood: "untared", a number only
A_rel	whole-blood reading after the tare. Normal = 0.95–1.05
C-score	genomic clustering of the departures (healthy ≈ 1). Development: no band yet
Withheld	what the build does not print, and why

Faults

symptom	cause	action
QUARANTINE_INCOMPLETE_MANIFEST	sex or age not declared	supply --sex and --age
"not the dominant cell"	neutrophils < 50 %	none: the fraction is reported and A is withheld by rule

symptom	cause	action
"untared: 0 same-slide reference arrays"	no references supplied	run the references, then pass them with --slide-ref-A
refusal "no frozen neutrophil floor for this platform"	the specimen is not EPIC	450K floor pending

Part 2. The chain, step by step

Source: doors/CHAIN_SEQUENCE.md

The chain, step by step - derived from the code

Generated by chain/build_chain_sequence.py from the AST of run_sample.py, conductor_v3.py and stage_q_iam_a.py. Do not edit by hand: re-run the generator. Every step below is a call the code actually makes, in the order it makes it. Chain v3 is the only engine; the class-floor engine (v2) was retired on 2026-10-03 and is archived privately.

The live path (chain v3) - 20 steps, in order

#	step	implemented in	what it does
1	step_0_1_idat_arrival	stage_0_intake.py	SOP section 0.1_idat_arrival
2	step_0_2_manifest_creation	stage_0_intake.py	SOP section 0.2_manifest_creation
3	step_0_3_integrity_hash	stage_0_intake.py	SOP section 0.3_integrity_hash
4	step_0_4_control_probe_validation	stage_0_intake.py	SOP section 0.4_control_probe_validation
5	step_0_5_detection_pvalue_qc	stage_0_intake.py	SOP section 0.5_detection_pvalue_qc
6	step_0_6_bead_count_qc	stage_0_intake.py	SOP section 0.6_bead_count_qc
7	step_0_7_call_rate	stage_0_intake.py	SOP section 0.7_call_rate
8	step_0_7b_platform_coverage	stage_0_intake.py	SOP section 0.7b_platform_coverage
9	step_0_8_sex_check	stage_0_intake.py	SOP section 0.8_sex_check
10	step_0_9_decision_gate	stage_0_intake.py	SOP section 0.9_decision_gate
11	Stage 1 - IDAT calibration	stage_1_idat_calibration.py	Stage 1 - IDAT calibration to beta (SOP Stage 1, steps 1.1-1.2 + 1.5) _called by run_sample.py when given --grn/--red; skipped when given --betas_
12	platform_refusal	conductor_v3.py	None when the specimen is EPIC v1; otherwise the refusal text
13	stage_m_isolated	conductor_v3.py	Isolated / sorted neutrophils against their own floor. The own-floor state is kept as state_own_floor; the reading's state is
14	stage_a_composition	conductor_v3.py	EPIC blood composition (blood_composition_EPIC_v1): 8 groups from Salas purified EPIC cells; markers exclude the neutrophil sites;
15	stage_m_blood	conductor_v3.py	Whole blood: Met-A = mean H(beta) / mean H(e) at the neutrophil sites, e = sum_g f_g mu_g (EPIC purified group profiles)
16	stage_t_tare	conductor_v3.py	Same-run tare against >= MIN_REFS healthy references of the same specimen type run the same way (same slide, else same batch)
17	stage_mc_cscore	conductor_v3.py	
18	stage_q_iam_a.pat_site_table	stage_q_iam_a.py	Per-site table (pos, opp_A, err_A, opp_B, err_B) from one.pat(.gz) file, pipeline loyfer_pat_v1. pos = 'chrom:CpG index' _runs when run_sample.py is given --pat or --site-table_

#	step	implemented in	what it does
19	stage_q_iam_a.read	stage_q_iam_a.py	site_table: per-site table (see module doc). cell: e.g. 'neutrophils'. pipeline: the read-level pipeline that produced the table_runs when run_sample.py is given --pat or --site-table_
20	Report	MethylPhys_Interface/repor t_v3.py	one self-contained HTML page plus the JSON bundle

Stage modules in chain/ that the live path does not call

module	what it implements	status
stage_2c_trace_detection.py	Stage 2c - is there any evidence of a trace class in this specimen?	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
stage_4_6_patient_cmb.py	Stage 4.6 — the patient's sky (PROC-CMB-01, 2026-09-21). Replaces the retired patient_brightness_comparison.py	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
stage_a_composition_v2.py	Stage A - composition, atlas v2 solver (deconv_v2). Not called by chain v3 (conductor_v3.stage_a_composition uses blood_composition_EPIC_v1.json). Frozen settings (commissioned as PROC-DECONV-V2-01; specimen cell sets 2026-09-30)	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)

The toolkit (chain/TOOLKIT.md) - built, not yet wired into v3

stage (SOP 2b)	name	module	status
3	Atlas deconvolution	chain/deconv_v2.py	toolkit: not yet wired into chain v3
3b	Trace-cell detection	chain/stage_2c_trace_detection.py	toolkit: not yet wired into chain v3
3c	Foreign-cell detection	chain/toolkit_foreign_detection.py	toolkit: not yet wired into chain v3
4	NILC component separation	chain/nilc_celltype_deconvolver.py	toolkit: not yet wired into chain v3
4	NILC helper: lineage splitter	chain/Lineage_Splitter/lineage_splitter.py	toolkit: not yet wired into chain v3
10	Directional decomposition	chain/Runtime Matrices/Directional Panel/bidirectional_decomposition.py	toolkit: not yet wired into chain v3
11	Sky map	chain/stage_4_6_patient_cmb.py	toolkit: not yet wired into chain v3
11b	Surface brightness	chain/toolkit_surface_brightness.py	toolkit: not yet wired into chain v3
12	Sky statistics	chain/sky_statistics.py	toolkit: not yet wired into chain v3
12b	Difference map	chain/serial_mode.py	toolkit: not yet wired into chain v3

Part 3. The toolkit

Source: chain/TOOLKIT.md

chain/TOOLKIT.md - the toolkit: built, kept, not yet wired into chain v3

Every module below is in the repository and importable, and **none of it is part of a reading**. Each enters chain v3 at commissioning, one stage at a time, after its own pre-registered check has passed on v3 and the outcome is recorded in doors/ (order and checks: . . /sop/MethylPhys_CPG_SOP_v3.md section 2b). Every module carries the line *Toolkit: not yet wired into chain v3; enters the chain at commissioning with its own pre-registered check*.

Stage numbers and names are those of SOP v3 section 2b. Paths are relative to

Biological_Physics/MethylPhys/. The class-era engine these modules were first used with (chain v2) was retired on 2026-10-03 and is archived privately; nothing here imports it.

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status		
3	Atlas deconvolution	chain/deconv_v2.py (DeconvV2); wrapper chain /stage_a_composition_v2.py; development copies atlas/v2/postbuild/scripts/deconv_v2.py, methylphys_v2.py	one joint non-negative solve over the atlas v2 cell types, $f \geq 0$, $\sum f = 1$, inverse-variance weights iterated, markers chosen per cell against its nearest cells; presence by bootstrap of the marker loci	the atlas v2 reference (parquet, built by atlas/v2/scripts; not stored in the repository), a beta vector on the Stage 1 scale; frozen solver settings in stage_a_composition_v2.py	not yet wired into v3		
3b	Trace-cell detection	chain/stage_2c_trace_detection.py; panel chain/Runtime Matrices/trace_detection_panel_v1.json	inverse-variance score test below the non-negativity boundary: fit without the candidate, project its profile off the fitted columns, regress the residual on it unconstrained	a beta vector; the panel (addresses, per-class means and SDs, thresholds)	not yet wired into v3. Retired from v2 on 2026-09-30 because its thresholds were a healthy-donor percentile and it read classes, not cells; restored as toolkit on 2026-10-03 (SOP 2b). Its line must be re-set without a population at commissioning		
3c	Foreign-cell detection	chain/toolkit_foreign_detection.py (stage_2d_foreign_detection); panel chain/Runtime Matrices/Celltype_Marker/detection_panel_v3.json	one joint NNLS of the panel markers on [blood reference columns, foreign-cell templates]; each template's fraction read against its measured noise floor; three or more above floor = "material not resolved"	a beta vector, the panel, optional composition-guard record	not yet wired into v3 (extracted 2026-10-03 from the retired conductor's Stage 2d)		

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status		
4	NILC component separation	chain/nilc_celltype_deconvolver.py (NILCCelltypeDeconvolver)	internal linear combination per cell type: variance-weighted generalised least squares over the markers, ridge-regularised, bootstrap CI per fraction	atlas CSV (atlas/IAMatlasREBUILD.csv.xz, decompressed), marker file chain/Runtime Matrices/Celltype_Marker/iamatlas_celltype_markers_v0_2.json, a beta vector	not yet wired into v3		
4	NILC helper: lineage splitter	chain/Lineage_Splitter/lineage_splitter.py (LineageSplitter)	splits one compartment's mass among its member classes by a small NNLS on the contrast CpGs (	mean_a - mean_b	>= min_delta, weighted by the contrast), with a condition-number check; reports the compartment only when the split is ill-conditioned	atlas CSV, a beta vector, the compartment fraction and its members	not yet wired into v3
10	Directional decomposition	chain/Runtime Matrices/Directional_Panel/bidirectional_decomposition.py; panel_directional_panels_v1_0.json beside it	per-CpG signed z against the panel's frozen direction, summed into a directional composite beside the pooled-entropy reading, so a departure's sign (toward disorder or toward over-order) is kept	a beta vector, the directional panel	not yet wired into v3		
11	Sky map	chain/stage_4_6_patient_cmb.py (patient_sky_sigma, render_plate); mapping chain/Runtime Matrices/Patient_CMB/iamatlas_cpg_to_healpix_nside128.npz built by chain/Runtime Matrices/Patient_CMB/build_healpix_mapping.py; earlier mapping atlas/healpix_mapping/	residual $r = \beta - \sum f \mu$ per CpG, scaled by the propagated atlas posterior and the array's SNP-probe noise, projected onto HEALPix NSIDE 128 (RING) in genomic order; panels masked below the presence floor; Mollweide plate	a beta vector, fractions, atlas means and SDs (atlas CSV), the mapping, chain/Runtime Matrices/Patient_CMB/presence_floors_v1.json	not yet wired into v3		

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status		
11b	Surface brightness	chain/toolkit_surface_brightness.py (attach_brightness_c i)	95 % interval on a reading by Monte Carlo through $A = H(\text{beta_mean}) / H_{\text{min}}$, using the atlas per-CpG brightness posterior SD at the scored loci	atlas/iamatlas_class_archives/*_REBUILD.tar.xz (per-CpG brightness CSVs), the scored loci and floor	not yet wired into v3 (extracted 2026-10-03 from the retired v1 clinical conductor; reads class-era fields until rewired)		
12	Sky statistics	chain/sky_statistics.py (masked_spectrum, bandpowers, shuffled_null); null suite chain/CPG_Null_Runner/cpg_null_runner.py; per-run tool registry chain/cmb_tools.py	masked pseudo angular power spectrum (healpy anafast) in six bands; spatially shuffled null (pixel values permuted within the mask); label-permutation, split, injection and look-elsewhere (Bonferroni, null N8) nulls; a registry that marks each borrowed tool PASS / FAIL / NOT_RUN / NOT_BUILT on a bundle	a HEALPix residual map (stage 11); for the null suite, a procedure's per-sample readings; for the registry, a run bundle	not yet wired into v3. Not built: look-elsewhere by simulation over the sky, apodised mask, beam smoothing, cell-type covariance in the separation (GLS), Fisher degeneracy of the composition, ILC on the residual sky, per-specimen composition posterior, cross-spectra between cell panels		
12b	Difference map	chain/serial_mode.py (check_same_person, delta_sky, delta_cells, trajectory)	per-address beta difference of two draws of one person on the addresses both measured, refused unless both draws share the identifier hash, array type and pipeline; per-cell A differences across draws (design doors /PROC_SERIAL_01_PREREG.md)	two runs (bundles and beta vectors) of one person	partly built: the per-address difference exists; the difference drawn as a sky map is not built. delta_cells and trajectory read class-era bundle fields (cells_all, tier) and must be rewired to v3 bundles		

Notes

- CPG_Null_Runner/cpg_null_runner.py null N7 loads the synthetic patient generator, which was retired with chain v2; N7 reports DEFERRED

until a v3 generator exists. The other nulls run without it.

- cmb_tools.py checks fields of the retired v2 bundle (patient_sky, foreign_detection, second_opinion); on a v3 bundle every entry reads

NOT_RUN until the stage it checks is wired in.

- The class-era wrappers that called these modules inside the retired conductor (the stage 4.6 sky wrapper, the class-level NILC

second opinion) were not carried over: they read retired files (class identity loci, beta scale maps). Each stage gets a v3 wrapper when it is commissioned.

Sources

manual/OM_v3_neutrophil_chain.md sha256 be036719f280

doors/CHAIN_SEQUENCE.md sha256 f617cc3ed7fd

chain/TOOLKIT.md sha256 cc09ad81fcda